

SUBHAM DANGAR

MSc in Data Science and AI, RKMVERI, Belur

+91 7029325530
dangarsubham@gmail.com

GitHub

LinkedIn

Portfolio



SUMMARY

M.Sc. Data Science student at RKMVERI with a strong mathematical foundation and hands-on experience in Machine Learning, Deep Reinforcement Learning, and distributed AI systems. B.Sc. Mathematics (Hons) graduate with CGPA 8.01. Built end-to-end ML pipelines, deployed APIs, and developed AI-powered semantic search systems independently.

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
MSc in Data Science, 2025–Present
CGPA – 6.3 (till sem 1)

Bankura Christian College
BSc (Honours) in Mathematics, 2022–25
CGPA – 8.01

Higher Secondary (XII), WBCHSE, 2022
94.8%

Secondary (X), WBBSE, 2020
92.14%

SKILLS

- Languages:** Python, C, C++, R, SQL
- ML/DL:** Supervised Learning, Deep Q-Networks, Semantic Search, Model Evaluation
- Frameworks:** NumPy, Pandas, Matplotlib, Scikit-Learn, sentence-transformers, Dask, FastAPI
- Tools:** SQLite, Git/GitHub, Jupyter, Google Colab, VS Code, Excel

COURSE WORK

- Machine Learning, Artificial Intelligence, Deep Learning, LLM, Agentic AI, Blockchain
- Probability and Statistics, Linear Algebra, Optimization, Time Series Analysis
- Distributed Computing, Data Structures and Algorithms

ACHIEVEMENTS & SCHOLARSHIP

- INSPIRE Scholarship – Govt. of India (Merit-based)

CERTIFICATES

- NPTEL – Python Programming
- NPTEL – C++ Programming

PROJECTS

Crop Disease Detection using Remote Sensing Images

Python, PyTorch, CNN, OpenCV, NumPy

Feb 2026 – Apr 2026

Developed a deep learning-based system for crop disease classification using remote sensing imagery. Applied image preprocessing and augmentation techniques and trained CNN models for automated disease detection. Evaluated model performance using accuracy, precision, recall, F1-score, and confusion matrix analysis.

Repository-Aware Autonomous Coding Agent

Python, Flask, Ollama, Sentence-Transformers

Mar 2026 – May 2026

Developed a fully local autonomous coding agent with a VS Code-like web IDE and an SWE-agent style execution loop for planning, code editing, testing, and self-correction. Implemented repository analysis, embedding-based retrieval, safe SEARCH/REPLACE editing, loop protection, and multi-language code execution using local LLMs via Ollama.

Distributed Educational Video Search System

RKMVERI

Mar 2026 - May 2026

Developed an AI-powered system to retrieve top 3 relevant YouTube timestamps from multilingual user queries. Used sentence-transformers and cosine similarity for semantic retrieval, ranked results by relevance and view count. Implemented parallel search with Dask, a semantic cache (SQLite), and a web fallback mechanism for reliability.

Spam Email Detection System

Python, scikit-learn, TF-IDF, Flask

Sep 2025 – Nov 2025

Built end-to-end ML pipeline with text preprocessing and TF-IDF feature extraction. Trained and compared Naive Bayes, Logistic Regression, and Random Forest models. Evaluated using precision, recall, F1-score, and ROC-AUC; deployed model via Flask for real-time prediction.

Mountain Car RL Agent (DQN)

Python, Gymnasium, Deep Q-Network

Sep 2025 – Nov 2025

Implemented Deep Q-Network with experience replay, target network, and epsilon-greedy exploration strategy. Tuned hyperparameters and tracked learning curves to achieve stable convergence in the Gymnasium environment.

ACADEMIC RESEARCH

A Study on Countability

B.Sc. Dissertation

Feb 2025 – Jun 2025

Studied finite, countable, and uncountable sets with formal proofs: Cantor's diagonal argument, countability of N, Z, Q .